

Cole Stevens

Squamish, BC
colestevens.ca

cole.stevens@hotmail.com
linkedin.com/in/cole-stevens-19288518b/

TECHNICAL SKILLS

Bare-metal embedded systems (ATmega, MSP430), serial communication (UART, I2C, RS-232, USB)
Software and hardware peripheral interfacing, UI for mechatronic systems, Unity game design

Proficient: C#, C/C++, HTML/CSS

Familiar: Python, Javascript, MATLAB, OpenCV, Tensorflow, Go, VBA, Assembly

ACADEMIC

Masters of Science in Computer Science | Georgia Institute of Technology, USA

Sept 2025 - Present

- Artificial Intelligence Specialization.
- Anticipated graduation: August 2027.

Bachelor of Applied Science, Mechanical Engineering | University of British Columbia, Canada

Graduated May 2021

- Mechatronics Specialization.
- Graduated with Distinction, achieving an 86% average.
- Completed cooperative education program.

TECHNICAL WORK EXPERIENCE

Front-End Web Developer (Part-time) | International Waters Governance

Remote | November 2025 – Present

- Migrating the organization website from Weebly to WordPress.
- Updating site content and structure.
- Creating new and adapting existing custom scripted HTML components.

Freelance Math/Engineering Expert | Scale AI

Remote | September 2024 – August 2025

- Trained generative AI models by prompting and evaluating math and engineering specific responses.

International Travel

March 2023 – May 2024

- Traveled throughout Europe and Asia; volunteered on farms and pursued rock climbing.
- Developed a video game in Unity and participated in a game development competition.

Associate Mechanical Engineer | STEMCELL Technologies

Hybrid, Vancouver BC | June 2021 – February 2023

- Designed, developed, and tested laboratory automation systems, sub-systems and consumables.
- Co-inventor: WO2024103186A1 – *Automated Fluid Exchange System*, STEMCELL Technologies, 2024
- Led development of bare-metal, multi-axis mechatronic and pneumatic manufacturing equipment, achieving a 50% improvement in product performance and reliability.
- Engineered custom mechatronic equipment that automated test procedures and data analysis, reducing development and production testing time by 80%.
- Developed low-level drivers for peripheral interfaces and optimized real-time machine cycles using hardware and timer interrupts to manage state.
- Collaborated on the design of a spinning blood filter for sample purification, focusing on the injection-molded rotor, filter, and enclosure design.
- Created Excel and SolidWorks macros, communication materials on microcontroller system design, and provided programming guidance for team members.
- Took the initiative of learning and applying Design of Experiments (DOE) data analysis techniques using JMP to enhance variable analysis efficiency, thereby improving testing rates and costs. Consolidated my research into an accessible resource for team members.

- Completed drawings, performed tolerance stack-up analysis, and applied GD&T principles across multiple prototype instruments.

Instrumentation Co-op | STEMCELL Technologies Inc.

In-person, Vancouver BC | January 2020 – March 2020

- Designed and built machines to automate laboratory procedures in Instrumentation R&D.
- Developed and conducted test procedures to aid development processes.
- (Laid off due to the pandemic, but rejoined full-time after graduation.)

Continuous Improvement and Transformation Co-op | Signify

In-person, Langley BC | May – December 2018

- Improved products and factory processes within the Continuous Improvement department.
- Developed an employee database using Microsoft Access and SQL.
- Automated workcenter instruction document creation with VBA to comply with ISO9001 standards.
- Communicated with managers, supervisors, and factory staff to implement efficiency improvements.

TECHNICAL PROJECTS

Unity Video Game Project

November 2023 – Present

- Competed in a game contest and made a simple fishing video game in two days.
- Continued development of the game independently into a top-down role playing game.
- Strengthened high-level programming skills, learned the Unity API, implemented shaders, and created original game art.

Interferometer Capstone Project, UBC

Vancouver BC | January 2018 – May 2018

- Partnered with Zaber Technologies to improve their interferometer resolution and measurement reliability at the nanometer scale.
- Studied the effects of air currents and temperature fluctuations on displacement.
- Designed a datum reference system and applied FEA and Fourier transform techniques to evaluate performance.

OTHER WORK EXPERIENCE

Bike Mechanic | StraightLine Bike and Ski

Fernie BC | May – August 2017

- Installed, replaced, and serviced brakes, suspension, pivot, and hydraulics on high end mountain bikes.

Floor Staff | StraightLine Bike and Ski

Fernie BC | November 2015 – May 2016

- Managed point-of-sale transactions and assisted customers with outdoor equipment.
- Performed ski tuning and maintenance.

Volunteer Soccer Coach | Rosedale Community

Calgary AB | April – July 2015

- Coached a mixed-gender soccer team (ages 14-17), leading practices and managing game strategies.